

Chapter 7: ACTIVITY: Joins Pivots dataRetrieval

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1 ACTIVITY: Joins Pivots dataRetrieval

This activity and the needed data are at https://github.com/VT-Hydroinformatics/7-Activity-Joins-Pivots_dataRetrieval

1.1 Load pacakges

Load the tidyverse, dataRetrieval, and patchwork packages. Set the theme for plots to theme_classic()

1.2 Problem 1 (8 pts)

Using `readWQPqw`, read all the chloride (00940) data for the New River at Radford (03171000). Select the `ActivityStartDate`, `MonitoringLocationIdentifier`, `CharacteristicName`, `ResultMeasureValue`, `ResultMeasure.MeasureUnitCode`, and `USGSPCode` columns. Use the `head()` function to print the beginning of the output from `readWQPqw`.

Call these data `newriverWQ`

1.3 Problem 2 (8 pts)

Using the `read_waterdata_daily` (daily values) function, download discharge (00060), temperature (00010), and specific conductance (00095) for the New River at Radford from 2007 to 2009 (regular year), making sure to `skipGeometry`. Keep the columns for `monitoring_location_id`, `parameter_code`, `time`, `value`, `unit_of_measure`. Use `head()` to show the beginning of the results of your download.

Call these data `newphys`

1.4 Problem 3 (6 pts)

Do a left join on `newphys` and `newriverWQ` to add the chloride data to the daily discharge, temp, and conductivity data. hint: you will join on the date. Preview your data below the chunk using `head()`.

Save these data as `newwqp`

1.5 Problem 4 (10 pts)

Create a line plot of Date (x) and Flow (y) using `newwqp`. You will need to filter the data based on parameter code! Create a scatter plot of Date (x) and chloride concentration (y). Put the graphs on top of each other using the `patchwork` library.

1.6 Problem 5 (4 pts)

Use `newwqp` to create a scatter plot of Specific Conductance (y) and Discharge (x). You will need to `pivot table`! Challenge: what could you do to get rid of the warning this plot generates about NAs.

1.7 Problem 6 (6 pts)

Read in the GG chem subset data and plot Mg__E1 (x) vs Ca__E1 (y) as points.

1.8 Problem 7 (10 pts)

We want to look at concentrations of each element in the #6 dataset along the stream (Distance), which is difficult in the current format. Pivot the data into a long format, the data from Ca, Mg, and Na __E1 columns should be pivoted. Make line plots of each element where y is the concentration and x is distance. Use `facet_wrap()` to create a separate plot for each element and use the “scales” argument of `facet_wrap` to allow each plot to have different y limits.